Fully Implicit ADER–DG Update for the 1D Shallow-Water System

1 Goal of the fully implicit ADER–DG step

We consider the one-dimensional shallow-water equations with fixed bathymetry

$$b = b(x),$$

written as

$$\partial_t \boldsymbol{U} + \partial_x \boldsymbol{F}(\boldsymbol{U}) = \boldsymbol{S}(\boldsymbol{U}, x), \tag{1}$$

with

$$\boldsymbol{U} = \begin{bmatrix} h \\ q \end{bmatrix}, \qquad \boldsymbol{F}(\boldsymbol{U}) = \begin{bmatrix} q \\ \dfrac{q^2}{h} + \dfrac{1}{2} g h^2 \end{bmatrix}, \qquad \boldsymbol{S}(\boldsymbol{U}, x) = \begin{bmatrix} 0 \\ -g h \, b_x \end{bmatrix}.$$

Assume that at time t^n the DG solution $\boldsymbol{U}_h^n(x)$ is known. The objective is to compute $\boldsymbol{U}_h^{n+1}(x)$ at $t^{n+1} = t^n + \Delta t$.

The fully implicit ADER–DG idea is to introduce, in each spatial cell K_i, an unknown polynomial $\boldsymbol{Q}_i(x,t)$ over the complete space–time slab

$$K_i \times [t^n, t^{n+1}].$$

Thus

$$\boxed{\boldsymbol{U}_h^n \quad \longrightarrow \quad \boldsymbol{Q}_h(x,t) \quad \longrightarrow \quad \boldsymbol{U}_h^{n+1}.}$$

2 Known solution and unknown space–time polynomial

Let

$$K_i = [x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}].$$

At time t^n, the known DG polynomial is

$$\boldsymbol{U}_h^n(x)\big|_{K_i} = \sum_{a=0}^{p} \widehat{\boldsymbol{U}}_{i,a}^n \phi_a(x). \tag{2}$$

Over $[t^n, t^{n+1}]$, introduce

$$\tau = \frac{t - t^n}{\Delta t} \in [0,1].$$

We seek

$$\boldsymbol{Q}_i(x,t) = \sum_{a=0}^{p} \sum_{\ell=0}^{r} \widehat{\boldsymbol{Q}}_{i,a\ell} \, \phi_a(x) \psi_\ell(\tau), \tag{3}$$

where ψ_ℓ are temporal basis functions. The vectors

$$\widehat{\boldsymbol{Q}}_{i,a\ell} = \begin{bmatrix} \widehat{h}_{i,a\ell} \\ \widehat{q}_{i,a\ell} \end{bmatrix} \in \mathbb{R}^2$$

are the unknowns of the time step.

It is convenient to use one combined space–time index $m = (a, \ell)$ and write

$$\theta_m(x,t) = \phi_a(x)\psi_\ell(\tau), \qquad \boldsymbol{Q}_i(x,t) = \sum_{m=1}^{N_{st}} \widehat{\boldsymbol{Q}}_{i,m}\theta_m(x,t),$$

with

$$N_{st} = (p+1)(r+1).$$

Therefore one cell contains $2N_{st}$ scalar unknown coefficients.

3 Space–time DG residual

For every space–time test function θ_k, we impose the shallow-water equations weakly on $K_i \times [t^n, t^{n+1}]$. After integration by parts in time and space, the vector residual is

$$\begin{aligned}
\boldsymbol{\mathcal{R}}_{i,k} = & \int_{K_i} \theta_k(x,t^{n+1})\boldsymbol{Q}_i(x,t^{n+1}) \,\mathrm{d}x \\
& - \int_{K_i} \theta_k(x,t^n)\boldsymbol{U}_h^n(x) \,\mathrm{d}x \\
& - \int_{t^n}^{t^{n+1}} \int_{K_i} (\partial_t\theta_k)\boldsymbol{Q}_i \,\mathrm{d}x \,\mathrm{d}t \\
& + \int_{t^n}^{t^{n+1}} \theta_k(x_{i+\frac{1}{2}},t)\widehat{\boldsymbol{F}}_{i+\frac{1}{2}}(t) \,\mathrm{d}t \\
& - \int_{t^n}^{t^{n+1}} \theta_k(x_{i-\frac{1}{2}},t)\widehat{\boldsymbol{F}}_{i-\frac{1}{2}}(t) \,\mathrm{d}t \\
& - \int_{t^n}^{t^{n+1}} \int_{K_i} (\partial_x\theta_k)\boldsymbol{F}(\boldsymbol{Q}_i) \,\mathrm{d}x \,\mathrm{d}t \\
& - \int_{t^n}^{t^{n+1}} \int_{K_i} \theta_k \boldsymbol{S}(\boldsymbol{Q}_i, x) \,\mathrm{d}x \,\mathrm{d}t. \qquad (4)
\end{aligned}$$

For each $k = 1, \ldots, N_{st}$,

$$\boldsymbol{\mathcal{R}}_{i,k} = \begin{bmatrix} \mathcal{R}_{i,k}^{(h)} \\ \mathcal{R}_{i,k}^{(q)} \end{bmatrix} \in \mathbb{R}^2.$$

The discrete equations are simply

$$\boxed{\boldsymbol{\mathcal{R}}_{i,k} = 0, \qquad k = 1, \ldots, N_{st}.} \qquad (5)$$

Thus the $2N_{st}$ scalar coefficients of $\boldsymbol{Q}_i$ are matched by $2N_{st}$ scalar residual equations in cell i.

4 How the algebraic residual is assembled

This is the essential step for implementing Newton's method.

1. Local vector of unknown coefficients

Collect all space–time coefficients in cell i into

$$\boldsymbol{c}_i = \begin{bmatrix} \widehat{\boldsymbol{Q}}_{i,1} \\ \widehat{\boldsymbol{Q}}_{i,2} \\ \vdots \\ \widehat{\boldsymbol{Q}}_{i,N_{st}} \end{bmatrix} \in \mathbb{R}^{2N_{st}}. \tag{6}$$

Here every entry $\widehat{\boldsymbol{Q}}_{i,m}$ is itself a two-component shallow-water vector.

For example, with $p = 2$ in space and $r = 2$ in time,

$$N_{st} = 3 \times 3 = 9,$$

so $\boldsymbol{Q}_i$ contains nine vector coefficients, i.e. 18 scalar unknowns in one cell.

2. Reconstruct $\boldsymbol{Q}_i$ from the current coefficients

For a given trial value of $\boldsymbol{c}_i$, reconstruct at every space–time quadrature point

$$\boldsymbol{Q}_i(x_\alpha, t_\beta) = \sum_{m=1}^{N_{st}} \widehat{\boldsymbol{Q}}_{i,m} \theta_m(x_\alpha, t_\beta).$$

Using these reconstructed states, evaluate

$$\boldsymbol{F}(\boldsymbol{Q}_i), \qquad \boldsymbol{S}(\boldsymbol{Q}_i, x),$$

and the two interface traces

$$\boldsymbol{Q}_i(x^+_{i-\frac{1}{2}}, t_\beta), \qquad \boldsymbol{Q}_i(x^-_{i+\frac{1}{2}}, t_\beta).$$

The required integrals in (4) are then evaluated by quadrature.

3. Couple the neighboring cells through the numerical flux

The interface fluxes are

$$\widehat{\boldsymbol{F}}_{i+\frac{1}{2}}(t) = \widehat{\boldsymbol{F}}\left(\boldsymbol{Q}_i(x^-_{i+\frac{1}{2}}, t), \boldsymbol{Q}_{i+1}(x^+_{i+\frac{1}{2}}, t)\right),$$
$$\widehat{\boldsymbol{F}}_{i-\frac{1}{2}}(t) = \widehat{\boldsymbol{F}}\left(\boldsymbol{Q}_{i-1}(x^-_{i-\frac{1}{2}}, t), \boldsymbol{Q}_i(x^+_{i-\frac{1}{2}}, t)\right).$$

Therefore

$$\boxed{\boldsymbol{\mathcal{R}}_i = \boldsymbol{\mathcal{R}}_i(\boldsymbol{c}_{i-1}, \boldsymbol{c}_i, \boldsymbol{c}_{i+1}; \boldsymbol{U}_h^n).}$$

The old solution $\boldsymbol{U}_h^n$ is known; the unknown arguments are only the coefficients of the space–time polynomials.

4. Stack the residual equations in cell i

Collect all vector residuals into one scalar algebraic vector,

$$\boldsymbol{r}_i = \begin{bmatrix} \boldsymbol{\mathcal{R}}_{i,1} \\ \boldsymbol{\mathcal{R}}_{i,2} \\ \vdots \\ \boldsymbol{\mathcal{R}}_{i,N_{st}} \end{bmatrix} \in \mathbb{R}^{2N_{st}}. \tag{7}$$

The condition for cell i is

$$\boldsymbol{r}_i(\boldsymbol{c}_{i-1}, \boldsymbol{c}_i, \boldsymbol{c}_{i+1}) = 0.$$

5. What exactly are $\boldsymbol{c}$ and $\boldsymbol{R}(\boldsymbol{c})$?

This point is important before writing the nonlinear system. The symbol q is reserved for the physical discharge $q = hu$. The bold symbol $\boldsymbol{c}$ denotes the algebraic unknown: the vector containing all coefficients of the space–time polynomials on the current time slab.

Let

$$N_{st} = (p+1)(r+1)$$

be the number of scalar space–time basis functions in one cell. Writing the vector coefficient associated with basis function θ_m as

$$\widehat{\boldsymbol{Q}}_{i,m} = \begin{bmatrix} \widehat{h}_{i,m} \\ \widehat{q}_{i,m} \end{bmatrix},$$

the unknown coefficient vector in cell i is, for example, ordered as

$$\boldsymbol{c}_i = \begin{bmatrix} \widehat{h}_{i,1} & \widehat{q}_{i,1} & \widehat{h}_{i,2} & \widehat{q}_{i,2} & \cdots & \widehat{h}_{i,N_{st}} & \widehat{q}_{i,N_{st}} \end{bmatrix}^T \in \mathbb{R}^{2N_{st}}.$$

Knowing $\boldsymbol{c}_i$ is equivalent to knowing the complete polynomial

$$\boldsymbol{Q}_i(x,t) = \sum_{m=1}^{N_{st}} \widehat{\boldsymbol{Q}}_{i,m} \theta_m(x,t)$$

over $K_i \times [t^n, t^{n+1}]$.

The global unknown is obtained by stacking the coefficient vectors of all cells:

$$\boxed{\boldsymbol{c} = \begin{bmatrix} \boldsymbol{c}_1 \\ \boldsymbol{c}_2 \\ \vdots \\ \boldsymbol{c}_N \end{bmatrix} \in \mathbb{R}^{2NN_{st}}.}$$

The known solution $\boldsymbol{U}_h^n$ is *not* part of $\boldsymbol{c}$; it enters the residual as known data through the initial-time term.

Now consider the residual. For every cell i and every space–time test function θ_k, equation (4) gives a two-component vector

$$\boldsymbol{\mathcal{R}}_{i,k}(\boldsymbol{c}) = \begin{bmatrix} \mathcal{R}_{i,k}^{(h)}(\boldsymbol{c}) \\ \mathcal{R}_{i,k}^{(q)}(\boldsymbol{c}) \end{bmatrix},$$

corresponding respectively to the mass equation and the momentum equation. These residuals depend on $\boldsymbol{c}$ because the states $\boldsymbol{Q}_i$, the physical fluxes, the source term, and the interface fluxes are all evaluated from the coefficients contained in $\boldsymbol{c}$.

For one cell, all test-function residuals are stacked into

$$\boldsymbol{r}_i(\boldsymbol{c}) = \begin{bmatrix} \boldsymbol{\mathcal{R}}_{i,1}(\boldsymbol{c}) \\ \boldsymbol{\mathcal{R}}_{i,2}(\boldsymbol{c}) \\ \vdots \\ \boldsymbol{\mathcal{R}}_{i,N_{st}}(\boldsymbol{c}) \end{bmatrix} \in \mathbb{R}^{2N_{st}}.$$

Finally, the global residual is

$$\boxed{\boldsymbol{R}(\boldsymbol{c}) = \begin{bmatrix} \boldsymbol{r}_1(\boldsymbol{c}) \\ \boldsymbol{r}_2(\boldsymbol{c}) \\ \vdots \\ \boldsymbol{r}_N(\boldsymbol{c}) \end{bmatrix} \in \mathbb{R}^{2NN_{st}}.}$$

Thus, $\boldsymbol{R}(\boldsymbol{c}) = \boldsymbol{0}$ means that, for every cell and every space–time test function, both the discrete mass residual and the discrete momentum residual vanish. For example, with $p = r = 2$, one has $N_{st} = 9$: each cell contains 18 scalar unknown coefficients and produces exactly 18 scalar residual equations.

The fully implicit ADER–DG step is therefore the nonlinear algebraic problem

$$\boxed{\boldsymbol{R}(\boldsymbol{c}) = 0.} \tag{8}$$

For the shallow-water system, $\boldsymbol{c}$ and $\boldsymbol{R}$ both have

$$2N(p+1)(r+1)$$

scalar entries.

5 Short Newton step

Because $\boldsymbol{F}(\boldsymbol{Q})$ and the numerical interface fluxes are nonlinear, (8) is nonlinear. Starting from an initial guess $\boldsymbol{c}^{(0)}$, Newton's method solves

$$\boxed{\boldsymbol{J}(\boldsymbol{c}^{(s)})\,\delta\boldsymbol{c}^{(s)} = -\boldsymbol{R}(\boldsymbol{c}^{(s)}), \qquad \boldsymbol{J} = \frac{\partial \boldsymbol{R}}{\partial \boldsymbol{c}},}$$

and updates $\boldsymbol{c}^{(s+1)} = \boldsymbol{c}^{(s)} + \delta\boldsymbol{c}^{(s)}$. The Jacobian is therefore differentiated with respect to the space–time coefficients $\widehat{\boldsymbol{Q}}_{i,m}$, not directly with respect to a separate $\boldsymbol{U}^{n+1}$. Because $\boldsymbol{r}_i$ depends only on cells $i-1, i, i+1$, the Jacobian has only the corresponding neighboring cell blocks.

6 How the updated solution U_h^{n+1} is obtained

After the nonlinear system has converged, all coefficients of $\boldsymbol{Q}_i(x,t)$ are known. The updated DG solution is the final-time trace

$$\boxed{\boldsymbol{U}_h^{n+1}(x)\big|_{K_i} = \boldsymbol{Q}_i(x, t^{n+1}).} \tag{9}$$

Using (3),

$$\boldsymbol{Q}_i(x, t^{n+1}) = \sum_{a=0}^{p} \sum_{\ell=0}^{r} \widehat{\boldsymbol{Q}}_{i,a\ell} \phi_a(x) \psi_\ell(1),$$

so the spatial DG coefficients at the new time are

$$\boxed{\widehat{\boldsymbol{U}}_{i,a}^{n+1} = \sum_{\ell=0}^{r} \widehat{\boldsymbol{Q}}_{i,a\ell} \psi_\ell(1).} \tag{10}$$

Thus $\boldsymbol{U}_h^{n+1}$ is unknown during the solve, but it is not an additional independent variable: it is already contained in the unknown space–time polynomial.

7 Complete update in compact form

known $\boldsymbol{U}_h^n$

$\Downarrow$

choose the space–time basis and unknown coefficients $\boldsymbol{c}$

$\Downarrow$

reconstruct $\boldsymbol{Q}_i$ and assemble all $\boldsymbol{\mathcal{R}}_{i,k}$

$\Downarrow$

$\boldsymbol{R}(\boldsymbol{c}) = 0$

$\Downarrow$

solve for the coefficients of all $\boldsymbol{Q}_i$

$\Downarrow$

$\boldsymbol{U}_h^{n+1} = \boldsymbol{Q}_h(\cdot, t^{n+1})$.

The central point is that the nonlinear variables are the coefficients of the unknown space–time polynomials. For every trial set of coefficients, the polynomials are reconstructed, the volume and interface terms are integrated to form the residual vector, and the coefficients are corrected until this residual vanishes. The final-time trace of the converged polynomial then gives the updated DG solution.